ROSE – InFIInit – intalnire 8, 03.02.2021

```
for i <- 0 to n-1 do
        for j <- i to n*n-1 do
                x <- x+1
```

T(n) = $\sum_{i=0}^{n-1}\sum_{j=i}^{n*n-1} 1 = \sum_{i=0}^{n-1}(n^2 - i) = \sum_{i=0}^{n-1} n^2 - \sum_{i=0}^{n-1} i = n^3 - \frac{n(n-1)}{2} = n^3 - \frac{n^2}{2} + \frac{n}{2}$

T(n) = O(n^3), T(n) = Ω(n^3)
T(n) = Θ(n^3)

T(n) = O(n^10) ? DA
T(n) = O(n^2)? NU

T(n) =Ω(n) ? DA
T(n)= Ω(n^10) ? NU

```
function fact(n)
begin
        if n == 0 then return 1
        else return n*fact(n-1)
end
```

T(n) = numarul de apeluri recursive in afara apelului initial
T(0) = 0
T(n) = 1 + T(n-1), n>=1

T(n) = n = Θ(n)

T(n) = numarul de apeluri fact(n) pentru inputul n
T(0) = 1
T(n) = 1 + T(n-1), n>=1

T(n) = n+1 = Θ(n)

```
function f(n)
begin
        if n <= 1 then return 1
        else return f(n-1) + f(n-2)
end
```

T(0) = 0
T(1) = 0
T(n) = 2 + T(n-1) + T(n-2) , n>=2 → T(n) = O(2^n)
T(n) <= 2 + 2T'(n-1) = T'(n) =

Rescriem T' ca T:

T(n) = 2 + 2 T(n-1)
2^1 T(n-1) = 2^2 + 2^2 T(n-2)
2^2 T(n-2) = 2^3 + 2^3 T(n-3)

2^(n-2) T(n – (n – 2)) = 2^(n-1) + 2^(n-1) T(1)

T(n) = 2^0 + 2^1 + 2^2 + … + 2^(n-1) -1 = 2^n – 1 – 1 = 2^n -2

<u>Dispersia interna – rezolvarea coliziunilor prin adresare deschisa</u>

- Nu avem spatiu suplimentar pentru legaturi.
- Se fac incercari sistematice pana cand se gaseste o locatie libera in T
  h: U x {0,1,…,m-1} → {0,1,…,m-1}
- Secventa de incercari trebuie sa fie o permutare a multimii de indici {0,1,…,m-1}
- Factorul de incarcare este subunitar! (n<=m).

+ **<u>Dispersia liniara:</u>** h(k,i) = (h(k,0) + i) mod m

<u>Exemplu</u>: m = 13, S.key = {18, 41, 22, 44, 59, 32, 31, 73}, h(k,0) = k mod m, |S| = n = 8

0	1	2	3	4	5	6	7	8	9	10	11	12
		41			18	44	59	32	22	31	73	

k	k mod 13	hash – index final
18	5	5
41	2	2
22	9	9
44	5	5,6
59	7	7
32	6	6,7,8
31	5	5,6,7,8,9,10
73	8	8,9,10,11